A Cracking Intersection

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A locksmith, a psychologist & a mathematician walk into a bar.
This is the result of their chat.

This little booklet makes up an exploration of the intersection between mathematics and psychology, with what hopes to be an interesting example. No prior knowledge is needed, but I will delve into the basics of combinatorics. By the end you will hopefully be entertained and look at the world from a slightly different perspective.

Contents

1 Where did this all begin?

To start right back at the very beginning would be to go back to Summer 2023, when I was walking down to work. But that feels like a stupid place to begin. So let's start at the end, the culmination of this little project, and work our way back to the beginning.

The start and end of any project is always quite boring to be honest. The start has a spark of inspiration that kicks everything into motion, and the end has a hint of tiredness, where you have exhausted every possible thought along the way. The most important, the juiciest, and the most interesting part is the middle, which I will very soon get into.

I'm currently writing this in a very nice café in Edinburgh, overlooking Calton Hill. After a few splendid days trekking around the Christmassy city, I thought that I should crack back on with some work. This brings us straight back to the end.

1.1 Psychology

I've been fascinated by human psychology for the best part of 5 years, from marketing & advertising, to body language & how we interact day-to-day. This mainly stemmed from my interest and passion in close-up magic, which is likely going to be the subject of inspiration for future issues of 'The Intersection'.

Psychology is the textbook for humans. It's how to understand and predict our behaviours, thought processes and actions. Basically it is the science behind our conscious mind. If you want to delve deeper and investigate some of this more, I highly recommend the text 'Decoded' by Phil Barden. It's marvellous.

1.2 Puzzles & Padlocks

Curiosity was an incredibly useful part of our lives about 1500-1000 years ago. Without it, we would never have evolved as much as we have now. We wouldn't understand Newtonian Gravity, Computers or Quantum Physics, although I still don't reckon that we understand that last one. Science would be incredibly stagnant without it. They say that curiosity killed the cat, and whilst I don't disagree that too much curiosity can be dangerous, a generous spoonful didn't hurt anyone. Our inherent curiosity is why the general public love puzzles, from wordsearches to sudokus, we can't get enough of them. I fall into that category of people and can often be found with my head buried in a puzzle book. In fact I've got a new one to explore this Christmas, but that's another matter. I digress.

Padlocks are intense puzzles to curious people. I'm sure that at some point in life you have marveled at locksmiths, or even been tempted to get yourself a lockpicking set. I know that I definitely have. Combination padlocks offer another challenge. They feel easy to crack but you can very clearly see and understand that the chance of you guessing the correct code is incredibly slim. This being said, we all try to crack it blindly the moment you get one in your hands. This booklet looks to increase your chances at blindly cracking a combination lock, whilst giving clear explanation as to why it works.

1.3 Mathematics

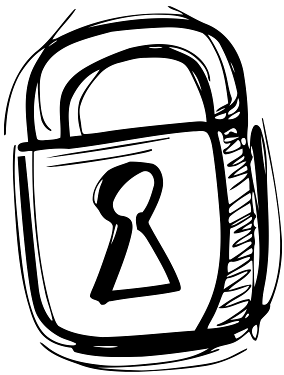
My background is in Mathematics and it is a field that I would like to continue studying and working in. I will present the exact figures and calculations to go along with these thoughts to hopefully aid your understanding. It is also why this booklet is laid out in the format of a set of Lecture Notes.

1.4 Recommendations & Resources

I would recommend following along with a pen and paper at any calculations, or at least allowing a moment for them to sink in. If you have a combination lock to hand, then that would be great too. I will be discussing both 3 & 4 digit padlocks, so either will suffice. Secondly, if you have never seen the topic of combinatorics before, then it may be worth also having a dice & coin to hand, but they are far from necessary. Finally, for those who want to learn more about combination lock puzzles, this Combination Lock Simulator is rather fun to play around with: 'https://scratch.mit.edu/projects/94462536/' All credit goes to 'Paddle2See' who created it on Scratch.

1.5 Let's Crack On

If I'm not mistaken, that should be everything to cover the set up of this project. If I have missed anything or you would like to get in touch with questions and thoughts, then shoot me a message on Instagram @samtomlin8.



2 A Brief Story

2.1 My Boss

Over the summer in question, I worked at a local family run chippie down by the coast. It is a lovely place, great food, great staff, very welcoming prices. Now my boss is a solid bloke, and also a very learned and intelligent man. He started off life as a car mechanic and then moved into catering, which he has now done for approximately 30 years. Throughout that time he has never stopped learning and is a very inquisitive person, so knows more about more things than most people probably know about particular topics. This being said, we've had chats about Quantum Theory, Mathematics, Cryptocurrency, and Current Affairs, to name just a few topics.

I've previously mentioned that I had the idea that is at the core of this project on my way to work. That being said, I mentioned it to my boss and he came out with this funny story.

2.2 After-School Fun

He said that when he was back at Secondary school, he would like to muck about with some of the other kids and have a laugh. This consisted of going down to the bike shed at lunch time, fiddling with the combination locks on 2 other kids bikes, unlocking them and then swapping. Then at the end of the day one kid would have to wait for the other to come out, so that they can swap locks and get their bikes back. When I asked how he unlocked it, he explained that with the old, not-so-good, combination locks you can feel which numbers are slightly looser & then easily play and crack it.

3 Combinatorics

3.1 Basics

Combinatorics, as the name suggests, is the branch of maths focused on things with finitely many possible different outcomes. For example, tossing a coin a certain amount of times, or rolling a dice multiple times. Each roll or toss is called an event. You can calculate the likelihood of each event happening by multiplying their probabilities. I'll run through it with this quick example for people who have a coin to hand, and then do a follow up example for those of you with a dice to hand.

3.2 Example: Coin

Let us work out the chance of getting a sequence: HEADS, TAILS, HEADS (HTH), if we toss the coin 3 times.

As previously mentioned, we count each toss of the coin as an event. This means that there are 2 outcomes from the first event; either H or T. Every event is completely independent, as tossing the coin once doesn't change the outcome of tossing the coin a second time. This implies that the second coin toss also has 2 possible outcomes H or T. Same applies to the third toss, with the endings being H or T.

Now we want to work out the chance of HTH. To do this we first calculate the total possible amount of different sequences. You could do this in a few different ways as the numbers are small, so I'll run through a couple.

The first is an exhaustive list: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.

The second is using a tree diagram, where you draw out each event, and the possible outcomes. Then to read it off, you read left to right and trace along one of the branches to come up with a possible sequence.

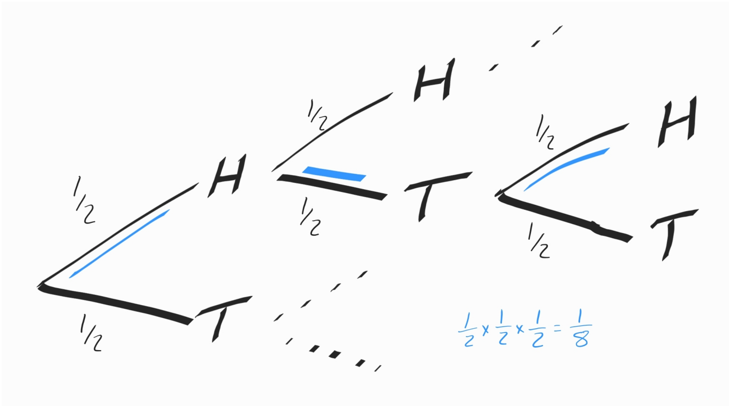


Figure: This is the tree diagram for tossing a coin 3 times. The blue lines show clearly the branches that we care about, for HTH. The rest of the tree has been omitted for clarity.

The final method, and the one that is more useful for general mathematics, is where combinatorics gets its name. You multiply the number of outcomes for each event. So in our case, event 1,2 & 3 each have 2 outcomes, so the maths becomes:

$$2 \times 2 \times 2 = 2^3 = 8$$

This means that we have 8 total outcomes. Now it is important to note that HTH is different to THH or even HHT, even though they contain the same outcomes, as the outcomes happen at different times. This means that the only outcome satisfying the question is HTH, hence only 1 outcome works. This implies that you get HTH, 1 out of 8 occasions.

In other words, the chance is $\frac{1}{8}$.

3.3 Example: Dice

Now let's consider the same style example but with a dice instead of a coin. Let's investigate the chances of rolling 325 on a normal 6 sided dice. That is rolling a 3 (Event 1), followed by rolling a 2 (Event 2) and then rolling a 5 (Event 3).

As this example follows on quite nicely from the previous one, I will skim over some of the details. If we start with Event 1, there are 6 possible outcomes. Same again for Events 2 & 3. Hence the total number of outcomes is:

$$6 \times 6 \times 6 = 6^3 = 216$$

Here I have used the combinatorial approach to calculating this as it is by far the quickest and simplest. A tree diagram works perfectly well, but the branches become fairly entangled and messy by the second roll. (As you will have 6 branches coming off of 6 previous branches, so about 42 total branches. 6 come from Event 1 and the other 36 from Event 2). The exhaustive method of writing out every sequence also works, but that would be awfully time consuming.

As with the previous problem, the order of the rolls is important to us. This means that 325, 352, 253, 235, 523, & 532 are all distinct. This will be a very important thing to us later in this project. Hence there is only 1 possible outcome that yields 325, so the chance is $\frac{1}{216}$.

3.4 Summary

I hope that this has provided a useful insight into basic combinatorics, or has jogged your memory back to school days when you might have learnt some of it. These are the key parts that you will need to understand going forward through this booklet. We are going to consider combinatorics in regards to 3 & 4 digit combination padlocks, which is why the order of the outcomes matters to us. Now all of the fiddly details have been thoroughly laid out, I shall set out the key idea.

4 How safe are Combination Locks?

4.1 3-Digit Locks

A 3-digit padlock is made up of the main body of the padlock and 3 metal turning wheels. Each wheel has the numbers 0-9 enscribed upon it. This means that each wheel has a possibility of 10 different numbers. To describe this in the same way as the examples, treat each wheel as an event, and each number on the wheel as an outcome. This means that we have 3 events & 10 outcomes from each event. That means that the total possible combinations for the padlock are:

$$10 \times 10 \times 10 = 10^3 = 1000$$

Imagine we have set the code for this padlock to 123, now there is a

$$\frac{1}{1000}$$

chance of cracking the padlock on a single guess. Pretty low if you ask me.

4.2 4-Digit Locks

Now a 4-digit padlock is much the same, but with a fourth metal wheel, and hence has a fourth event. This gives a total of 4 events & 10 outcomes from each event, meaning that the total combinations for the padlock are:

$$10 \times 10 \times 10 \times 10 = 10^4 = 10,000$$

This time, imagine that we've set the code to 1234, meaning that there now is a

$$\frac{1}{10,000}$$

chance of cracking the padlock on a single guess. Even slimmer chance.

4.3 Why isn't it quite as secure as we think?

This is where the human psychology comes into play and decreases the security of these significantly. When unlocking a padlock, you have to enter the correct code, and then the padlock will automatically open up. When you go to re-lock it, the correct code will of course be displayed, so you naturally change all of the digits showing, to some random digits. And this is where it goes down hill.

Take the first digit for example, we know that it is a 1, so we want to move the wheel from 1 so it is not easily guessable. We naturally try to hide this piece of information to make it harder to crack. Except, of course, no-one else is privvy to this bit of info. So they don't know to suspect the number 1, just because the first metal wheel has been left on a 1. What I'm trying to say, is that if the first digit of the code is 1, then it is highly likely that when it is rescrambled, it will not be left on a 1. The same logic can be applied to each of the other metal wheels.

Taking that into consideration, if we approach a 3-digit lock without knowing the code, it is safe to assume that none of the wheels display the correct digit. This is the key assumption and is at the crux of this idea. In trying to make the padlock safer, you in fact allow the person a chance to make an assumption, which reduces the safety considerably.

Now consider this as a lock with 3 wheels still but each only has 9 possible outcomes. The total possible combinations for this padlock are:

$$9 \times 9 \times 9 = 9^3 = 729$$

All of a sudden we have lost 271 potential combinations, all because you want to hide the digits. When you consider a 4 wheel version, the reduction in potential combinations is even more striking:

$$9 \times 9 \times 9 \times 9 = 9^4 = 6561$$

This time we have lost even more possibilities.

4.4 A Few Compromises

The next thing that we should probably consider is the best way to combat this. If we leave all digits of the code unchanged when scrambling, then of course nothing will happen, and the padlock will remain unlocked. If however you scramble one digit, but leave the rest unchanged, then you have:

$$9 \times 10 \times 10 \times 10 = 9000$$

possible different codes for a 4 digit padlock. Whilst this is very good, I reckon it is slightly too open. In particular, if by some pure chance or luck, the person fiddles around with only one wheel, they have a 1 in 4 chance of picking the right wheel and finding the code by pure trial and error.

I think the perfect method is to scramble only 2 of the wheels. This way it is a 1 in 6 chance of them picking the correct 2 wheels to fiddle with, and it also still has a fairly high amount of options for possible different codes:

$$9 \times 9 \times 10 \times 10 = 9^2 \times 10^2 = 8100$$

One final thought: the full security in this relies on nobody knowing how you scramble the padlock. If you choose, 1 wheel, 2 wheels or even 3 wheels to mix, the strength lies in the randomness of how you choose the method and also the prior knowledge of what method you have used to scramble it. Without that, it becomes even harder to crack, especially if you don't just scramble in a standard way.

5 Final Words

And that brings this little booklet to a close. I intended for it to be a short quick read about an idea that I thought was fairly abstract and interesting. On top of that, it has also provided the chance to showcase an intersection between 2 previously unlinked topics: Psychology & Mathematics. I plan on continuing the theme of random but interesting intersections over the next few issues, with the plan to widen your perspective and view on the world and the things in it. Hopefully the next time you see a combination lock, you think to yourself about the psychology that lies behind the security of it. Or at the very least you take away this thought:

Nothing is quite as disconnected as it seems.